PH.45320 **FALL 2026**

Static Cosmology

Lecture 2: Redshift and Medium Extinction

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Overview

Spectral redshift and line broadening are derived from transport through the medium rather than from scale-factor evolution.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$d \ln \nu / ds = -\eta_\nu(\rho_m, T_m)$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$1 + z = exp(\int \eta_\nu ds)$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\sigma_{line}^2 = \int D_\nu ds$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 2
- Extinction Atlas §1
- CMT redshift block Z-01–Z-09